Brice Fernandes

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Profile

Technology leader with a record of delivering in regulated, high-trust environments: application platforms for global banks, critical national infrastructure, and now learning platforms used by children in the UK, EU and US. Builds engineering organisations that treat security, data protection and compliance as part of how products are made, gives teams clear direction and real authority to decide, and sets AI strategy from first-hand experience of building AI systems. Helped define GitOps at Weaveworks, since adopted as an open standard under the Cloud Native Computing Foundation. An educator as well as an engineer: created a credit-bearing programming course taken by over 100,000 students, and helped shape the University of Cambridge Judge Business School's startup accelerator as one of its founding coaches.

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Experience

Director of Engineering at Raspberry Pi Foundation September 2024 - Present

Engineering leader for the Digital Products division of the Raspberry Pi Foundation, an educational charity whose mission is to enable all young people to realise their full potential through the power of computing and AI. Accountable for engineering across multiple product teams building the learning platforms used by young people and educators worldwide: around 20 engineers, led by five Engineering Managers. Report to the Chief Technology and Product Officer. Own engineering vision and strategy, org design and resourcing, the engineering budget, and the security, compliance and data-protection posture of the digital estate. With technology and product under one executive, also carry Foundation-wide technology work: the technology risk register and business continuity, the Foundation's data map, AI direction and open source strategy. Member of the Foundation's Leadership Team, representing engineering to the Executive team.

- Led the technical data residency strategy for a public-sector opportunity of ∼1M students, authoring the costed brief that framed the executive decision
- Directed engineering AI adoption through an adoption plan, a dedicated working group, and an adoption-maturity reporting tool built with external partners
- Built the Foundation-wide data map (∼90 systems, 20 data categories, 25 data flows) as the evidence base for ROPA and ongoing privacy compliance
- Led technical compliance for products used by children in the UK, EU and US: Online Safety Act assessments, digital age of consent, data protection, and the vendor security assessment for major US school districts
- Led the security programme for Digital Products: Cyber Essentials Plus certification (achieved August 2026), penetration testing, cloud access audits, software approval, business continuity and incident response
- Established a working-group operating model with charters, tiers and explicit decision rights: autonomous groups covering Security, Architecture & Infrastructure, AI Adoption, Accessibility, Design System, and Risk & Compliance, each with delegated authority and budget
- Designed and landed a technology-agnostic engineering career framework (Apprentice to Principal, plus a distinct management track), now in active use across 1:1s, reviews and progression decisions
- Grew and stabilised the team through recruitment at every level, contractor procurement, and structured investment in people development

KEYWORDS: *Engineering Leadership, Technology Strategy, Engineering Strategy, Org Design, Budget Ownership, Delivery Management, Architecture, AI Adoption, Responsible AI, Data Engineering, Data Protection, GDPR, ROPA, Online Safety, Data Residency, Compliance, Cyber Essentials Plus, Security, Career Frameworks, People Development, Recruitment, Open Source, Education, Charity, Non-profit*

DevOps Practice Leader at Capgemini January 2022 - June 2024

Senior Leader of the DevOps practice for Capgemini UK Digital Customer Experience (DCX), a practice of 30 to 40 people. Worked on commercial proposition development, aligning product, service and capability with clients, market units, and other delivery areas in DCX and beyond. Owned the technology radar and external thought leadership for the DevOps practice within the DCX delivery unit. Delivered client engagements for some of the UK's largest organisations, including Critical National Infrastructure in the energy sector.

- Worked on all aspects of capability management: products and offers, culture, training, recruitment and the career framework
- Senior technical leader on client engagements, the largest worth over £10M and delivered by consortiums, owning technical architecture and ways of working
- Managed four to five managers directly, and mentored other senior leaders
- UK point of contact for Capgemini's Global DevOps Community
- Security cleared to work on Critical National Infrastructure

KEYWORDS: *Senior Leadership, DevOps, Capability Management, Architecture, Delivery Management, Digital Transformation, Career Frameworks, Training, Recruitment, Product Management, Critical National Infrastructure*

Senior Solution Architect at Weaveworks September 2018 - September 2021

Architected, developed and delivered hybrid cloud application platforms for some of the company's most high-profile clients, including in multiple regulated industries. Responsible for technical leadership and client relationships across commercial engagements, from presales to delivery management, and for the design and delivery of GitOps and Cloud Native training for clients and partners.

- Helped define the principles of GitOps, since adopted as an open standard under the Cloud Native Computing Foundation, and taught them to clients, partners and the wider community
- Provided architecture consultancy for internal application platforms to global banks
- Led the design and architecture of the internal application platform to support the autonomous vehicle project of one of the world's largest companies
- Led the initial design and architecture workshop for the re-platforming effort of a large European telecommunications provider
- Trusted key-holder for critical internal systems
- Designed and implemented the customer experience playbook

KEYWORDS: *GitOps, Customer Experience, Delivery management, Kubernetes, Cloud Native, Presales engineering, Monitoring, Observability, Digital transformation, SRE, Training, Recruitment*

Senior Development Engineer at Weaveworks September 2017 - September 2018

Developed Go tools for managing Kubernetes at an open source company whose products were built in the open with their user communities. Worked on Cortex, the horizontally scalable Prometheus service, and contributed to Flux and Weave Net. Helped define the GitOps methodology and communicate it to the developer community. Presented GitOps at conferences in the UK and internationally and delivered training on Kubernetes and GitOps. Member of the on-call engineering team directly responsible for the uptime and maintenance of the company's online SaaS platform.

KEYWORDS: *Open Source, Kubernetes, GitOps, Cloud Native, Prometheus, Terraform, Go, On-Call, Incident response, DevOps, SRE, Developer Experience, Training, Recruitment*

Founder at Decacoder October 2015 - January 2017

Founded an online education company teaching developers through a custom-built platform. Led development of the media delivery platform and its continuous deployment pipeline on AWS, and was responsible for the educational design and student experience alongside every other area of the business. Began building Decabot, an adaptive learning AI project, which won an Innovate UK grant. Served as the company's registered Data Protection Officer.

KEYWORDS: *Education, Founder, Product development, Leadership, Educational design, UX, Data protection, JavaScript, React, AWS, Continuous integration, Continuous deployment*

Software Engineer, Cambridge Broadband March 2011 - October 2013
Developed software for a microwave backhaul product: cross-platform graphical applications, embedded C network management over SNMP, and internal tools. Member of the process improvement team that redesigned the company's software and product development process. Took responsibility for recruiting C developers and for selecting and managing contractors.
KEYWORDS: *Embedded, C, Python, Linux, SNMP, Agile, Process improvement, Recruitment, Contractor management*

Teaching & Mentoring

Instructor, Unity3D Development Course August 2014 - August 2015
Designed and delivered an online course teaching programming and the fundamentals of computer science to novices through game development in C# and Unity3D. Funded through Kickstarter and delivered on Udemy, the course reached over 100,000 students and was accredited for 10 European Credit Transfer System credits (∼10% of an undergraduate degree).
KEYWORDS: *Education, Curriculum design, Computer science, Online learning, Game Development, C#, Unity3D, Kickstarter*

Startup Mentor & Founding Coach, Accelerate Cambridge 2013 - 2016
Selected as a founding coach and mentor for the University of Cambridge Judge Business School startup accelerator Accelerate Cambridge, and helped define the shape of the programme. Delivered a startup founder education curriculum and coached teams on product design, technology strategy, development process, team management, pitching for investment and clear communication. Several of the startups mentored received prestigious awards, including winning the Pitch@Palace competition.
KEYWORDS: *Startups, Coaching, Business modelling, Pitching, Leadership, Investment, Technology Strategy*

Instructor & Coach, Ignite Programme July 2015, July 2017
Instructor and coach on the University of Cambridge Judge Business School's Ignite programme, a one-week intensive course for early-stage enterprises. Led workshops that helped attendees apply the teaching to their own businesses.
KEYWORDS: *Startups, Education, Coaching, Leadership, Event facilitation, Business Modelling*

Founder, Cambridge Programmer Study Group July 2014 - December 2016
Founded the most active programming meetup in Cambridge, meeting twice a week to study computer science topics. Over two years the group studied Machine Learning, Cryptography, Concurrency and Functional Programming, amongst other topics.
KEYWORDS: *Community building, Computer science education, Machine Learning, Neural Networks, Cryptography, Concurrency, Functional Programming, Compilers*

Independent Work

factor-q
A self-hosted, event-driven runtime for operating multi-agent AI systems on long-running projects. Every agent action is an event on a durable bus, giving a full audit trail; each agent can use a different model provider, tools run sandboxed, and spending limits are enforced. Written in Rust on NATS, with the source available under the Business Source License.
KEYWORDS: *AI Agents, Multi-agent Systems, Agent Orchestration, Event Sourcing, Rust, NATS*

narrate
An open source tool that turns a Markdown script and a folder of pictures and clips into a narrated video on the author's own computer, with the voice-over generated by a local speech model and every scene timed to its narration. Runs as a command line, web studio, HTTP API, Docker image or GitHub Action.
KEYWORDS: *Text-to-Speech, Video Production, Python, FFmpeg, Docker, Open Source*

TaskFlow
An open source Kanban task tracker in which people and AI agents work on the same boards, with configurable workflow state machines and an audit trail of every action. Reached through a command line, terminal interface, HTTP API or Model Context Protocol server.
KEYWORDS: *AI Agents, Model Context Protocol, MCP, Human-AI Collaboration, Go, Open Source*

Codren
A fully interactive, domain-specific AI experience built on Claude: a persistent character for a tabletop role-playing group. It remembers each person across sessions, learns actively by finding the gaps in its own knowledge and asking about them, and personalises how it responds to each participant. Grounded in a searchable archive, with access enforced on the server by role so that material sealed from a player never reaches the model.
KEYWORDS: *Conversational AI, Agent Memory, Active Learning, Personalisation, RAG, Claude*

Skills

LEADERSHIP: Engineering strategy and vision, Org design and resourcing, Budget ownership, Technical recruitment, Career frameworks and progression, People development and coaching, Delivery and operations, Incident response and business continuity, Curriculum design and technical training.

GOVERNANCE & COMPLIANCE: Data protection (GDPR, ROPA, former Data Protection Officer), Online safety, Security programme ownership, Cyber Essentials Plus, Vendor security assessment, Data residency, Risk management, Policy and audit.

AI & DATA: AI strategy and adoption, Multi-agent systems and agent orchestration, Model Context Protocol (MCP), Retrieval-augmented generation, LLM-assisted development workflows, Adaptive learning, Machine learning, Natural language processing, Agent-based modelling, Data mapping and governance.

TECHNOLOGY: Cloud native architecture, GitOps, Kubernetes, AWS, GCP, Terraform, Site Reliability Engineering, Observability, Python, Go, Rust, JavaScript, C, Functional programming, Test Driven Development.

Certification & Training

Santa Fe Institute Introduction to Agent Based Modelling	Achieved 2024
Scrum Alliance Certified Agile Leadership 2	Achieved 2024
Scrum Alliance Certified Agile Leadership 1	Achieved 2023
AWS Certified Machine Learning - Specialty	Achieved 2022
AWS Certified Security - Specialty	Achieved 2022
MIT Media Lab: AI-Generated Media Course	Achieved 2021
AWS Certified Solutions Architect - Professional	Achieved 2020
AWS Certified DevOps Engineer - Professional	Achieved 2020

Selected Talks

Embedded Test Driven Development (slides, code)	March 2024
Site Reliability Engineering Best Practices (video)	October 2020
Bret Fisher's Docker and DevOps Podcast (video)	November 2019
Delivering Quality at Speed with GitOps (video)	August 2019
Tutorial: Hands-on GitOps, KubeCon (video, slides, code)	December 2018
Building Objects With Functions (interactive version, recording)	June 2015

Publications

The Evolution of Complementary Cognition: Humans Cooperatively Adapt and Evolve through a System of Collective Cognitive Search.
Taylor, H., Fernandes, B., & Wraight, S. (2021). Cambridge Archaeological Journal, 1-17.
doi: 10.1017/S0959774321000329

Education

BSc Physics with Computer Science, The University of Sheffield — 2006 - 2010